REGENCY SALMA / ZOBAY NEIGHBOURHOOD

Complete Technical Architecture, Stack Breakdown & Domain Hierarchy Specification

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1. Executive Summary & Vision

Regency Salma (commercialized as Zobay Neighbourhood) is an enterprise residential community operating system. Unlike legacy gated community management tools that treat flat ownership and user accounts as simple strings or single-role entries, Regency Salma was engineered around a core architectural axiom: **Identity ≠ Authorization**. A single real-world human being (User) can simultaneously be an **Owner** of Unit 9A, a **Tenant/Resident** of Unit 4B, and an elected member of the **Leadership Council**. Every permission, pass, gate access check, and chat privilege is dynamically derived from verified multi-table relationships rather than a static role flag.

The project was recovered from a legacy prototype into a mathematically strict, deterministic architecture featuring **34 relational database tables**, zero un-migrated schema mutations (strict *migrate deploy*, no *db push*), strict token-versioning security, full offline/online state reconciliation, and multi-role workflow support across mobile and web interfaces.

2. Complete Technology Stack & Purpose Matrix

Every library and technology in Regency Salma was selected to enforce determinism, security, and low-latency mobile/web responsiveness:

Technology	Category	Exact Role & Architectural Purpose in Project
Expo Router 57	Mobile / Web Framework	File-based routing for universal apps. Powers simultaneous native deployment (iOS/Android) and zero-config Web application execution via metro-bundler.
React Native 0.86 & React 19	Frontend Core	Cross-platform mobile UI runtime delivering 60fps animations, native haptic integration, safe-area boundary management, and responsive layouts across phones, tablets, and desktop browsers.
Node.js 20+ & Express 5	Backend API	High-throughput RESTful routing and middleware pipeline with async error handling, custom security cascades, rate limiting, and structured JSON responses.
Prisma ORM 7.9+ / 7.10	Data Access Layer	Type-safe database ORM managing relational constraints, atomic transactions ($transaction), and strictly tracked SQL migrations (prisma/migrations).

PostgreSQL 16	Relational DB	ACID-compliant primary datastore with 34 relational tables, multi-column uniqueness indexes, foreign keys, and audit logging for security gate passes.
JWT & TokenVersion	Authentication	Dual-token auth (Access + Refresh). Includes database-backed tokenVersion to allow instant global session revocation when credentials or roles change.
TanStack React Query 5	State Management	Server-state caching, background revalidation, optimistic mutations, and automated cache invalidation upon gate check-ins or visitor approvals.
Zod 4	Validation	Runtime schema enforcement on API payloads (e.g., verifying mandatory purpose on visitor requests and 6-digit numeric OTP formatting).
WebSockets & Polling	Real-Time Comms	Bi-directional real-time chat, security gate alerts, and instant resident notification for arriving guests and courier deliveries.
Docker Containerization	Infrastructure	Isolated, reproducible execution of PostgreSQL with volume persistence and environmental portability across dev, staging, and production.

3. Architectural Layers & Data Flow

The system follows a layered, decoupled service architecture ensuring that gate physical operations cannot be compromised by application UI states:

LAYER 1: CLIENT PRESENTATION (EXPO & REACT NATIVE WEB)
• **Resident App:** Home Dashboard, QuickActions Radial, Household Members, Visitor Pass Creation, Community Spaces, Help-Desk Tickets.
• **Workforce / Guard Portal:** Gate scanner, OTP verification, PermanentPass fast-track check-in, Visitor Departure log.
• **Management & Leadership:** Tenancy approvals, Personnel attendance, Community broadcast announcements, Role approvals.

LAYER 2: NETWORK & RECONCILIATION LAYER
• REST API endpoints over HTTPS with strict Authorization: Bearer <token> headers.
• Token refresh rotation & active session reconciliation every 60 seconds.
• Express Rate Limiters guarding authentication endpoints against credential stuffing and brute force.

LAYER 3: BUSINESS DOMAIN & CONTROLLERS (EXPRESS 5)
• **Auth Controller:** Strict registration flow (Public manager registration requires Leadership sign-off).
• **Household Controller:** 28-day validity lifecycle, cascade pause/revocation transactions.
• **Visitor Controller:** Concurrency locks on party sizes, gate arrival validation, 6-digit cryptographic OTP generation.

LAYER 4: PERSISTENCE & ACCESS CONTROL (PRISMA & POSTGRESQL 16)
• 34 Relational Tables with foreign key enforcement.
• Atomic transactions preventing race conditions during physical admission at community gates.

4. Comprehensive Domain Hierarchy & Stakeholder Model

A critical strength of Regency Salma is how it categorizes people, legal rights, and physical access. The hierarchy branches cleanly between **Legal Property Rights**, **Physical Residency Rights**, **Administrative Authority**, and **Operational Gate Access**.

Entity / Role	Relationship / Scope	Privileges & Concrete Operational Capabilities
LEADERSHIP (Council / RWA / Board)	Community-wide Authority	• Supreme administrative authority over the entire community. • Reviews and approves/rejects new **Manager** and **Security Staff** appointments. • Publishes community-wide Broadcast Announcements (URGENT / NORMAL). • Strict database guard (requireLeadership); cannot be spoofed by manager or resident accounts.
PROPERTY OWNER / BUYER (UnitOwnership)	Unit-level Legal Ownership	• Holds registered deed / ownership record (UnitOwnership ACTIVE) for one or more physical units. • Can reside in the flat (Residency OWNER ACTIVE) or be an absentee/non-resident landlord. • Approves or rejects incoming **Tenants** attempting to register for their unit. • Retains legal control over permanent access grants assigned to the property.
TENANT (UnitResidency: TENANT)	Unit-level Leased Residency	• Rents a unit from an Owner. Initial registration creates a PENDING residency. • Once approved by the Owner or Community Manager, gains active residential rights. • Can generate Guest/Delivery OTPs, register personal vehicles, and book community amenities. • Does NOT have rights to approve other tenants or transfer unit ownership.
RESIDENT / FAMILY MEMBER (HouseholdMember)	Unit-level Occupant	• Family members, co-residents living within an active owner or tenant household. • Can request visitor passes, view gate notifications, and engage in unit/community chat. • Can have their access state set to ACTIVE, PAUSED, or REVOKED.
COMMUNITY MANAGER (Manager)	Operational Oversight	• Dispatches and oversees help-desk maintenance tickets (Plumbing, Electrical, Infrastructure). • Assigns tasks to maintenance personnel and verifies completion. • Manages staff rosters, duty schedules, and vendor registrations.
WORKFORCE & SECURITY (Watchman / Guard)	Perimeter & Gate Checkpoint	• Stationed at physical entrance gates. • Validates 6-digit visitor OTPs and checks in pre-approved delivery/cab drivers. • Scans **PermanentPass** QR codes for recurring household staff (maids, cooks, drivers). • Immediate rejection if household member is PAUSED or REVOKED.

VISITORS & DELIVERY (Visitor)	Transient / Timed Access	• **Guests:** Receive a 6-digit OTP; locked after 5 failed attempts. • **Deliveries / Cabs:** Pre-approved by resident; direct gate clearance without OTP. • **Unannounced Visitors:** Guard enters details → push notification to resident → resident one-tap approve/reject.

5. Critical Security Invariants & Audit Proofs

Regency Salma incorporates strict mathematical invariants tested and verified at the database and endpoint layers:

1. Cascade Revocation & Pausing Invariant

When an Owner or Resident marks a HouseholdMember (e.g. domestic maid or driver) as PAUSED or REVOKED, an atomic Prisma transaction immediately updates all linked PermanentPass records to PAUSED/REVOKED. If that individual attempts gate check-in, the gate API returns an immediate 400 Bad Request error. Access cannot leak due to caching.

2. Renewal Guard Invariant

A PermanentPass can only be renewed for an additional 28-day window if its associated parent HouseholdMember is currently ACTIVE. Any attempt to renew a pass for a PAUSED or REVOKED member triggers a strict rejection.

3. Strict Leadership Isolation

Administrative community endpoints are protected by requireLeadership. Even a valid Community Manager or Owner receives a 403 Forbidden on leadership-only routes.

4. Public Registration Trapdoor

Users attempting to register publicly with the role of MANAGER are placed in a quarantined PENDING status. They have zero managerial privileges until an active Leadership member executes a formal approval.

5. Mandatory Purpose Enforcement

All visitor creation requests must supply a validated, non-empty purpose string (validated via Zod). Anonymous visitor passes are rejected at the schema level.

6. Cross-Community Invisibility (404 Invariant)

To prevent information leakage across multi-society deployments, queries across community boundaries do not return 403 Forbidden; they return 404 Not Found, keeping foreign resources completely invisible.

6. Visitor Lifecycle State Machine

The gate validation engine tracks visitors through an audited, deterministic state transition cycle:

Lifecycle Stage	Trigger / Event	Security Checks & State Outcome
CREATED	Resident creates pass in app	Zod validation checks purpose, unit residency, and generates 6-digit OTP.
PENDING	Visitor arrives unannounced	Guard inputs visitor phone; real-time alert sent to resident device for approval.
APPROVED	Resident approves pass	Gate status flips to APPROVED; ready for admission.
PARTIALLY_ADMITTED	Party size admission (e.g. 2 of 5)	Transactional check: admittedCount < partySize to prevent over-admission.
CHECKED_IN	Guard verifies OTP or Pass	Timestamp recorded; visitor physically admitted to community premises.
CHECKED_OUT	Visitor departs through gate	Security logs exit time; pass transitions to completed state.
EXPIRED / REVOKED	TTL expires or resident cancels	Pass becomes invalid; subsequent gate check-in attempts return DENIED.

7. Relational Database Architecture (34 Core Tables)

The PostgreSQL schema managed by Prisma encompasses 34 specialized tables grouped into five logical subsystems:

Subsystem	Key Relational Tables Managed
Identity & Membership Core	User, Membership, RefreshToken, AuditLog
Community & Property Topology	Community, Block, Unit, UnitOwnership, UnitResidency
Household & Access Management	HouseholdMember, PermanentPass, HouseholdVehicle, HouseholdPet, HouseholdHelp
Visitor & Gate Operations	Visitor, VisitorLog, GateCheckIn, GatePassAudit
Workforce, Operations & Help-Desk	Personnel, Department, Attendance, Ticket, Task
Social, Communication & Amenities	Channel, ChannelMember, Message, Space, SpaceMessage, Announcement, Poll, PollOption, PollVote

8. Production & Local Deployment Runbook

Reproducible Startup Commands:

• **Database (Docker):** docker run --name regency-salma-test-db -e POSTGRES_USER=regency_test -e POSTGRES_PASSWORD=postgres -e POSTGRES_DB=regency_salma_test -p 5435:5432 -d postgres:16

• **Backend Migrations:** npx prisma migrate deploy (Strict: never use *db push* in production).

• **Seed Demo Accounts:** npx ts-node src/seed.ts

• **Launch Backend:** npm run dev (Runs Express on port 3005).

• **Launch Frontend:** npx expo start --web (Runs Expo Metro on port 8081).